Devdatta Kasture

devdatta.kasture@gmail.com | (309) 251-9298 | linkedin.com/in/devdattakasture/ | https://github.com/TheCrickMaster

EDUCATION

University of Illinois Urbana-Champaign **Urbana, IL**
Bachelor of Science in Computational Mathematics; Data Optimization *Expected Graduation: May 2026*
Minor in Computer Science
GPA: 3.70 / 4.00

PROFESSIONAL WORKING EXPERIENCE

Country Financial **Bloomington, IL**
Data Engineering Intern *May 2023 – Aug 2023*
- Utilized Apache Spark framework to automate daily data ingestion into existing data pipelines
- Programmed complex query design in Cloudera environment for large scale data analysis
- Engineered automatic data validation application utilized PySpark, NumPy, Pandas, and SQL
- Observed and promoted proper Agile methodology to develop, test, validate, and produce software solutions given a deadline
- Collaborated on case study to implement AI technology on insurance and financial services within the company in collaboration with three colleagues

Relevant Skills: Python (NumPy, Pandas), Git Microsoft Azure, SQL, Tableau, Microsoft Office Suite

PERSONAL PROJECTS

Ameritegra Trading Systems *In Production*
- Engineering an algorithmic trading application utilizing KDB+/q, Python, and C++ workflows
- Designed table schema for Historical Database in KDB+ for automated data ingestion of equity and options data
- Generated qSQL queries to effectively manage and analyze time series data
- Produced PyKx Script to leverage python functionality to generate real-time q process for data ingestion in Real Time Database

Relevant Skills: KDB+/q, Tick Data Architecture, Git, Linux, Python (Numpy, Pandas, PyKx), qSQL, IPC, Pulse Visualization Software

Market Fragmentation Analysis on Multiple Liquidity Sources *May 2025*
- Engineered a market fragmentation tool to analyze real-time market data for alternative trading strategies in KDB+/q and Python
- Utilized qpython and selenium to generate synchronous and asynchronous queries for data aggregation
- Constructed a low-level q process for real time market data integration on remote servers
- Produced parameter specific scripts for multi-market aggregation using KDB+/q built in parsing functionality
- Utilized Pulse data visualization software to craft a performance benchmarking companion

Relevant Skills: KDB+/q, Python, Tick Data Architecture, Git, Linux Selenium, Pulse Visualization Software, Low-Level Systems

Genomic Sequencing Predictive Modelling Analysis *Feb 2025*
- Developed a genomic sequencing classifier to predict disease outbreaks in C++
- Constructed well-defined functions implementing programming principles of inheritance and polymorphism
- Deployed Valgrind to detect memory leaks and optimization dynamic memory allocation
- Migrated software module to Linux based machine to maximize resource and memory efficiency
- Derived stochastic algorithm utilizing simulated annealing to predict likely causes of viral outbreaks

Relevant Skills: C++, Valgrind, Low-Level Systems, Git, Linux, Linear Programming, Stochastic Analysis

PROFESSIONAL DEVELOPMENT AND LEADERSHIP

Society for Industrial and Applied Mathematics (SIAM) Student Chapter **Urbana, IL**
- Serving as society member for UIUC SIAM Chapter *Aug 2024 – Present*
- Collaborating with other members to further promote research in applied mathematics
- Learned about applied mathematical applications for life sciences, engineering sciences, computer sciences, economics, and various other disciplines through seminar and guest speakers

ADDITIONAL INFORMATION

Certifications: Bloomberg Markets Concepts (BMC), KX KDB+/q Developer Level 1 Certification
Volunteer Experience: Washington Helps Its People Winter Food Drive (2018, 2019, 2020, 2022), Chinmaya Mission Peoria Aim for Seva Dhanam (2019, 2022)
Other Skills: Java, Marathi (Intermediate), French (Intermediate),
Interests: Computer Programming, Game Theory, Sports Data Analytics, Environmental Initiatives, Self-Sustaining Gardening, Music Theory, Music Engineering